



03/09/21  
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# COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD  
P.M.B 1023, OTA, OGUN STATE, NIGERIA

## B.Sc DEGREE EXAMINATION

COLLEGE: SCIENCE AND TECHNOLOGY

SESSION: 2020/2021

COURSE CODE: CSC227

COURSE TITLE: COMPUTER ARCH. AND ORGAN. I

DEPARTMENT: CIS

SEMESTER: OMEGA

CREDIT UNIT: 2

TIME: 2 HOURS

**INSTRUCTION: ANSWER QUESTION 1 AND ANY OTHER 2 QUESTIONS**

### Question One (30 marks)

- a. Minicomputers in the eighties had 18 address signals and 18-bit address bus.  
Answer the following questions: **3mks**

- What was the address space of these computers?
- What may have been the largest possible memory of these computers in bytes if the memory location is 1 Byte?
- How long should the program counter (PC) of these computers be?

- b. i) What is pipelining? **2mks**  
ii) Calculate how long it will take to execute on a regular **non-pipelined and pipelined** architecture. There are 4 operations with 5 stages each: IF, ID, EX, M and WB; stage1= 8ns, stage2 = 9ns, stage3 = 11ns, stage4 = 13ns, stage5 =16ns. Provide the Gantt Charts for both pipeline and non-pipeline operations. Also calculate the Speedup of the Pipeline system **6mks**

- c. i) Describe the three types of data hazards. **1½mks**  
ii) What type of data hazards do the following instructions belong? **1½ mks**

- MULTD F0, F2, F4 *write*  
DIVD F10, F0, F6 *read*
- DIVD F10, F0, F6 *read*  
ADDD F6, F8, F2 *write*
- MULTD F0, F2, F4  
SD F8, 50 (R5)

- d. Find the minimum sum-of-products expression represented by the K-map in Figure 1. 'd' means use either 1 or 0 as convenient: 2mks

|       |       |    |    |    |    |
|-------|-------|----|----|----|----|
|       | $x_1$ | 00 | 01 | 11 | 10 |
| $x_3$ | 0     | 1  | 0  | 1  | 1  |
|       | 1     | d  | 0  | 1  | d  |

Figure 1

- e. Using Boolean algebra, produce a simpler circuit from the circuit given in Figure 2. 4mks

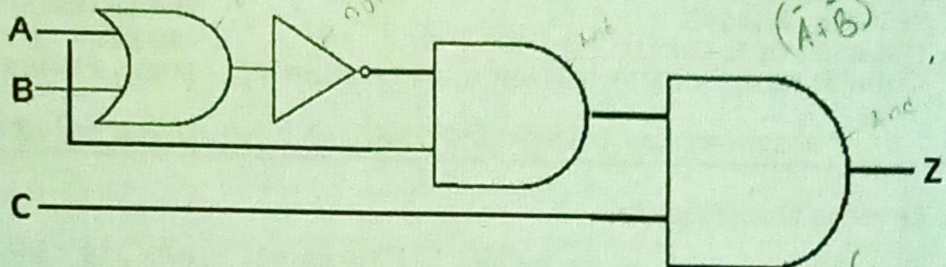


Figure 2

- f. i) What is Instruction Set Architecture?  
 ii) Use a zero address, one address and two address instruction format to write a Pentium 4 assembly language program segment for the expression below. For each format calculate the number of bytes required given that opcode is 2 bytes and memory access is one byte. 1mk

$$Z = V * ((U + W)/(X * Y));$$

9mks

### Question Two (20marks)

- a. Write down a signed decimal number with a value of -25 in 8-bit fixed point presentation in each of the three 8-bit modes for the representation of signed numbers. 3mks
- b. i) Write the value of 4.75 in 32-bit floating point format IEEE 754. 4mks  
 ii) The floating point value is written in the memory as 44FAC000h. Which value does it represent if it is written in the format IEEE 754? 4mks  
 iii) What do you understand by Instruction Format and Instruction Set? 2mks
- c. i) State the difference between Von Neumann Architecture and Harvard Architecture (include diagrams) 3mks  
 ii) What do you understand by Von Neumann Bottleneck? Suggest three ways to mitigate this bottleneck. 4mks

### Question Three

a. Consider Table 1

- i) Describe each Addressing Mode 1½mks
- ii) Represent the instruction given in a memory segment for each mode using the values provided. 4½mks
- iii) For each Addressing Mode also determine the **ADDRESS** and **VALUE** of the operand that is added to register **R1**. 3mks

Table 1

| s/n | Memory Address (initial) | Instruction           | Addressing Mode   | Remark                          |
|-----|--------------------------|-----------------------|-------------------|---------------------------------|
| i   | 0x017A                   | Add R1, #0xC21A       | Immediate         | -                               |
| ii  | 0x017A                   | Add R1, [R5]          | Register Indirect | R5 = 0xA75F, operand is 0x1BEC  |
| iii | 0x017A                   | Add R1, [PC + 0xC21A] | PC Relative       | R5 = 0x01700, operand is 0x1BCD |

b. Complete the Table 2. Show all necessary work done to arrive at each answer. Use **10bits**. 6mks

Table 2

| Value        | Unsigned | S-M | 1's Complement | 2's Complement |
|--------------|----------|-----|----------------|----------------|
| 234          |          |     |                |                |
| -106         |          |     |                |                |
| 1111 0110 11 |          |     |                |                |

c. Use the reverse of the IEEE Single Precision to determine the value (Base 10) of the following hexadecimal (base 16) value: **BE7C2002H** 5mks

### Question Four (20 marks)

- a. Describe three categories of Instructions that we have. Give two examples of each. 6mks
- b.
  - i) State Four (4) clear differences between RISC and CISC. Give two examples of processors in each category. 4mks
  - ii) Discuss the characteristics of computer memory in terms of the following:
    - 1) Access Methods
    - 2) Memory Hierarchy (Include diagram where necessary) 4mks
- c. ii) Determine the range of **POSITIVE** and **NEGATIVE** numbers that can be represented in a 16-bit register using **Sign-Magnitude**, **1's complement** and **2's complement**. 6mks

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**Question Five (20marks)**

- a. i) A processor can support a maximum memory of 4GB where the memory is word addressable (a word = 2 bytes). Determine the number of bits required to uniquely address the entire memory. **2mks**
- ii) Describe **End-Around Carry in 1's Complement?** Illustrate this using **153 - 142** in an 10-bit register. **4mks**
- b. i) List four (4) differences between DRAM and SRAM? **4mks**
- ii) What will be displayed (in binary and decimal) as the sum of 214 and 65 using an 8-bit register? Show details of work? **3mks**
- c. Consider a 4 stage pipeline processor. The number of cycles needed by the four instructions T1, T2, T3, T4 in stages S1, S2, S3, S4 is shown below:

|    | S1 | S2 | S3 | S4 |
|----|----|----|----|----|
| T1 | 2  | 1  | 1  | 1  |
| T2 | 2  | 3  | 2  | 2  |
| T3 | 2  | 1  | 1  | 3  |
| T4 | 1  | 2  | 2  | 2  |

- i) Draw the chart one iteration of the loop given below **6mks**
- ii) Determine the number of cycles needed to execute the following loop **1mk**
- for ( $j = 1$  to 2)(T1; T2; T3; T4;)